1. Purpose:

To stipulate the required tests for wound drainage set to ensure good quality products before releasing to the market.

2. Scope:

This procedure is applicable for the wound drainage set delivered sterile by Saudi mais company.

3. Definition:

Wound Drainage Set: A medical device used to remove waste fluid from the body is response to tissue damage from the patient wound.

4. Responsibilities:

4.1 Quality Controller is responsible for sampling & testing.

4.2 Quality supervisor is responsible for checking the obtained results.

4.3 Machine operators and assembly peoples are responsible to produced good quantity and good quality products.

5. Procedure:

5.1 Receiving Raw material:

- Ensure the received material from supplier is correct material in order to specification
- Inspect the material bag condition, must be no damaged at all bags.
- Take samples & inspect in order to AQL procedure, product description, specification with apply the AQL sampling plans for normal inspection.
- Record into OBM inspection report.

5.1.1 Inspection & test procedure for Raw Material:

Work Description	Requirements	Method of Check
Clarity / Appearance	Clear / Good Appearance	Visual
Condition of Bag	Good / No damage bags	Visual
Material Grades / Description	As per P. O	Visual / Physical
Appearance of granules	Good / Transparent / Colored	Visual
Granule's formation	No lobes formation	Visual
Lot number printed status	Clear printed	Visual
Arrangement of materials bags	Should be specified clear label	Visual

5.2 Injection Machine Procedure:

- Production controller will responsible to issue raw material and production order for injection Moulding for wound drainage components.
- Maintenance department will responsible to set the required mold with standard parameters.
- Mahine operator will responsible to feed required raw material as per production order.
- Quality controller will take sample as per AQL sampling plan and inspect the components as per required specification.
- Every 2 hours quality controller will check the components and make sure

5.3 Bellow Pump

- Quality controller responsible to make sure that bellow pump should be accurate in capacity, no damaged, no holes, no crack and should be leakage free.

5.4 Bellow pump cap

- Quality controller responsible to make sure that bellow pump cap should be smooth surface no, flashing, no black spots, no crack and damaged and free from irregularities.

5.5 Bellow pump Adaptor

- Quality controller responsible to make sure that bellow pump adaptor should be smooth surface, accurate in dimension and should be free from black spots and blockage.

5.6 Adaptor Cover

- Quality controller responsible to make sure that adaptor cover should be smooth surface, accurate in dimension, and should be free from damaged and crack.

5.7 Slide clamp

- Quality controller responsible to make sure that slide clamp should be smooth surface, accurate in dimension as per specification and should be free from black spots, damaged and crack.

5.8 Luer tip connector

- Quality controller responsible to make sure that luer tip connector should be transparent, accurate in dimension and should be free from damaged and crack.

5.9 Plastic Bend

- Quality controller responsible to make sure that plastic bend should be flexible, accurate in dimension and should be free from damaged.

- Machine operators are responsible to arranged good components in plastic bag with inspection tag QC stamp must be stamp on inspection tag to be confirm acceptable before transfer to production store area.
- The responsible person and machine operators must be carrying the good output products and with attached inspection tag, QC approval.

6. Extrusion Machine Procedure:

- Production controller will responsible to issue raw material and production order for wound drainage tube, redon drain tube and multichannel tube.
- Maintenance department will responsible to set the required die with standard parameters.
- Mahine operator will responsible to feed required raw material as per production order.
- Machine operators are responsible to submit initial sample to quality controller for inspection.

Quality controllers are responsible to make sure that wound drainage tube, redon drain tube and multichannel tubes are accurate in dimension ID /OD and length as per specification.

6.1 Wound drainage tube

- Quality controllers are responsible to make sure that tube should be smooth surface, accurate in dimension with and should be free from fish eyes, blackspots, holes and dust particles.

6.2 Redon Drain tube

- Quality controllers are responsible to make sure that redon drain tube should be smooth surface with dark blue X-ray line, clear printed and visible, with drainage holes, and should be free from flashing, black spots, and holes blockage.

6.3 Silicon multichannel tube

- Quality controllers are responsible to make sure that silicon multichannel tube should be silicon with X-ray line blue or white, accurate in dimension, with channels, and should be free from black spots, burn marks, and free from irregularities.

7. Receiving Bulk Products.

- Ensure the received bulk product are same as per our purchase order and specification
- Inspect the carton condition, must be in good condition, no damaged at all cartons with products.
- Take samples & inspect in order to AQL procedure, for normal inspection.

Controlled Copy

Hi vacuum Wound Drainage

Hi vacuum wound drainage should be accurate in capacity with tube and trocar, and strong gluing of components and should be free from damaged and leakage.

8. Receiving Bulk component.

- Ensure the received component same as per our purchase order and specification
- Inspect the carton condition, must be in good condition, no damaged at all cartons with products.
- Take samples & inspect in order to AQL procedure, product description, specification with apply the AQL sampling plans for normal inspection.

Trocar

- Trocar should be stainless steel, accurate in dimension, free from corrosion and rust particle.
- Record into OBM inspection report.

8. Assembly procedure

- Production controllers are responsible to issue all required components for assembly of wound drainage set.
- Each component of wound drainage set should be assembled by gluing as per diagram.
- Quality controller will responsible to take samples as per AQL sampling plan from final assembled products and make sure each components should be strong gluing and no pullout.
- Trocar assembled with tubing it should be strong no pull out.

8.1.1 In-Process Control for Wound Drainage set:

Work Description	Requirements	Method of Check
Tube / Bellow pump clarity	Clear / Good	Visual
Tube gluing / Insertion	Strong, No pullout	Visual & Physical
Tube with slide clamp	Available	Visual & physical
Fitting's status	Good fitting	Visual
Leak test for Bellow pump	No leakage Manual	Visual
Capacity (400ml)	200 ml, 400 ml, 600 ml	By water
Connector type	According to size	Visual
Trocar	According to size	Visual
Trocar sharpness	Sharp without damage	Visual & Physical
Redon tube length	500 mm with multi holes	By measurement

Bellow pump Graduation scale	Clear marking	Visual
Punching holes quality	w/o fibers, blockage & roughness	Visual
Connector fitting with Bellow pump	Strong without pull out	Visual & Physical
Correct specification	As per requirement	Visual & Physical
Hole drilled on bellow neck	Clear without fiber	Visual
Apply rubber on hole	Yes / or No	Visual
Hanger pieces	Available	Visual

9. Visual Inspection.

- When examined by normal or corrected to normal vision, under and minimum 2,5x magnification the external surface of the effective length of the drainage catheter shall appear free from extraneous matters, process and surface defects.
- The tube surface must be smooth, no bumps, no bubbles and no fish eyes at all.
- The tube murphy eyes must be smooth & round edge.
- Connector and tubing must be good fitting, no pullout.
- The tubing length should be accurate as per our required specification and size.
- The Trocar must be sharp and protected with cover.
- The bellow pump must be clean, clear, free from irregularities flashing, holes and blockage etc.
- Bellow pump fitting with connector must be strong no pull out.
- The connector and hanging tie, must be strong with the tubing no crack, no short shot.
- All connector fitting with tubes it should be strong and no pull out.
- All the components and accessories must be match and complete before going to packaging machine.

10. Quality Inspection

10.1 Corrosion resistance of trocar

When tested in accordance with the standard ISO20697 Annex B Trocar should be free from corrosion and rusts.

10.2 Resistance to deformation by suction

When tasted in accordance with the method given in standard ISO20697 Annex C. components should not deformed and detached during suction.

Controlled Copy

10.3 Bonding strength:

When tested in accordance with the method given in standard ISO20697 Annex E Product should be complies with the requirement of standard force.

10.4 Trocar:

Trocar should be with sharp tip, pure stainless steel and should be free from rust and corrosion.

10.5 Tube Detectability

The effective length of tube should be printed with x-ray line and X-ray line should be complies with the requirement of standard ASTMF640.

11. Packaging and Labeling:

- After issue the product for packaging, quality controller and production controller must be check before transfer to the packaging section.
- Each product package in blister pack or pouch and, the label contains the following information.
- Product name and code no., reference no., size, lot no., type, temperature, logo, sterilization logos, production date, expiry date, etc.
- During packaging process, quality controller will take samples as per AQL sampling plan and make sure package sealing, peeling, sealing width, forming, cutting and printing.
- After packaging on machine or by pouch, the product should be checked by the repacking personnel, packaging should be intact &in good condition.
- Then to pack in specified small boxes, and arrange in big carton.
- The big carton must be with label and detail information.

12. Sterilization:

- The finished products are sterilized as per work instruction (WI-7-5-1-1).

13. Final Inspection and Product Release:

- The Quality controller must be check, the unloaded products after sterilization.
- The quality assurance manager or quality control manager have to review the completed cycle parameters and inspection reports both documents have to be sign.
- After confirmation all the tests from the Lab, to release the sterilized products, the document once again has to review the quality control manager then sign.
- The production department responsible to record the quantity of products before transferring to the finished goods ware-house and then it could be released to the customer.

Controlled Copy

14. Sampling:

- Inspection & Test report for Raw material : F-823-T1
- Inspection and test report for trocar : F-823-T18
- In-process control for Wound Drainage : F-823-i12
- In-process control for packaging machine and final checking : F-823-i4-2
- Final inspection and Release Record : F-8-2-4

15. Reference:

ISO20697

16. Modification / Documentation change History:

Date	Changes	Rev.
23.11.2018	Item 4.3, 5.1.1,5.5 & 8 new added item 5.2,5.3,5.6, &9 modified	2
03.05.2021	Item 5.3 to 5.9, 6.1 to 6.3,7.1,9& 10 new added	3

Controlled Copy